

Knowledge Graphs

Lecture 5 - Knowledge Graph Applications

5.4 Knowledge Graph Programming

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Knowledge Graphs

Lecture 5: Knowledge Graph Applications

5.1 Ontologies in Action

5.2 Knowledge Graphs

5.3 RDF and OWL Knowledge Graphs

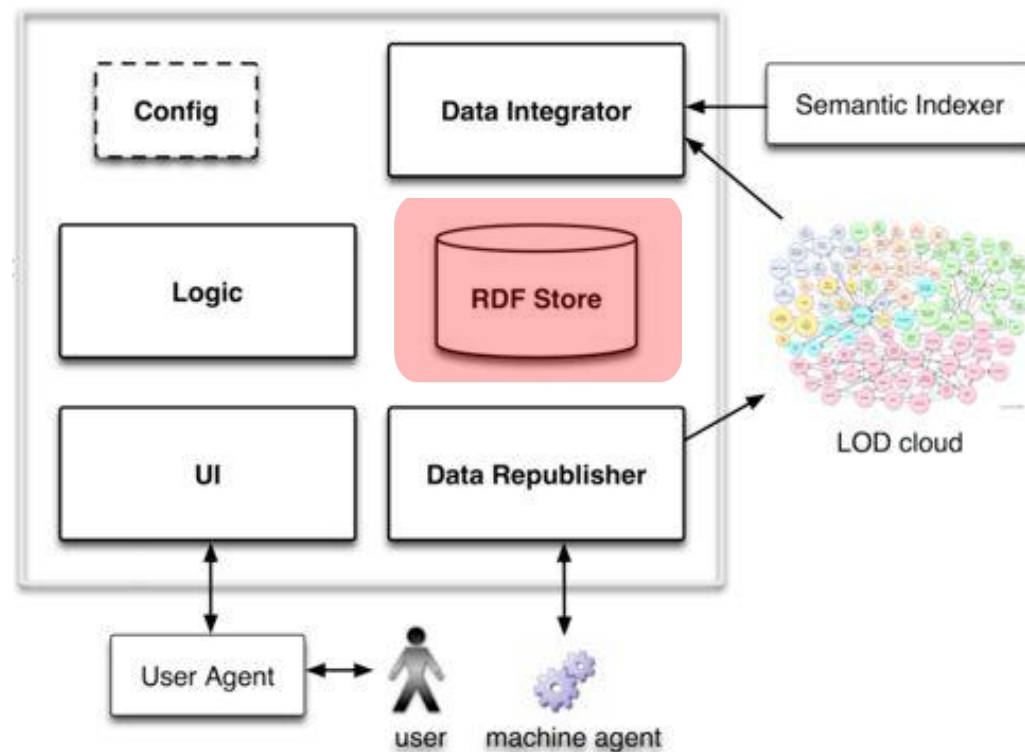
5.4 Knowledge Graph Programming

5.5 Knowledge Graph Visualization

5.6 Knowledge Graph Analytics

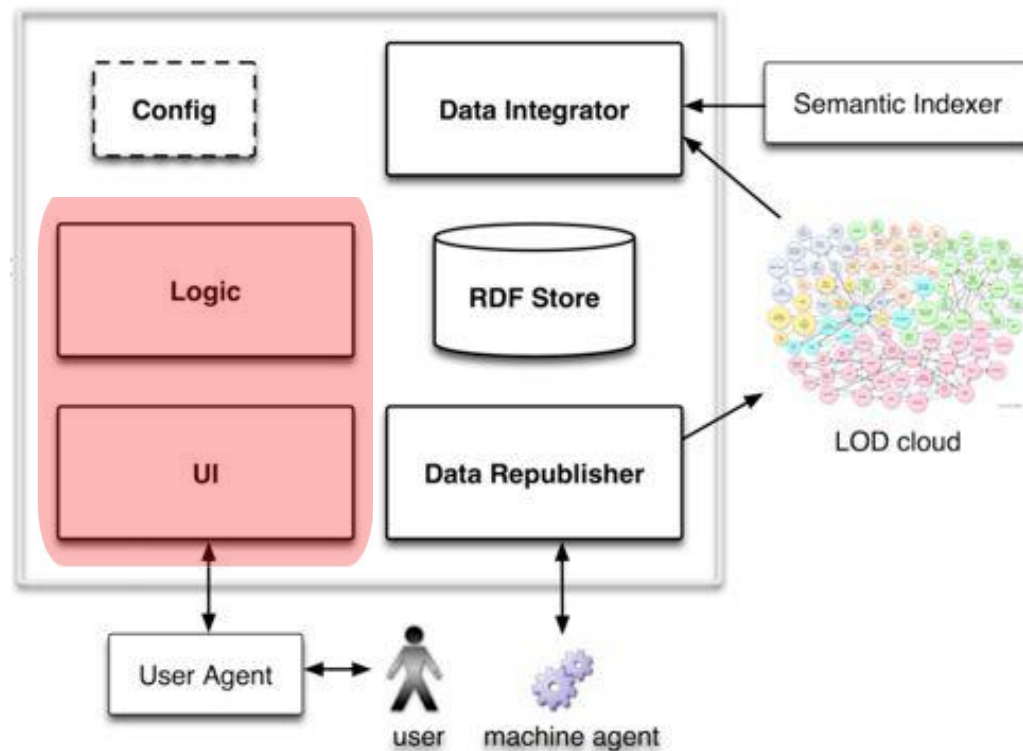
Knowledge Graph Driven (Web) Applications

- Required Components:
 - **Local RDF Store**
 - caching of results
 - permanent storage



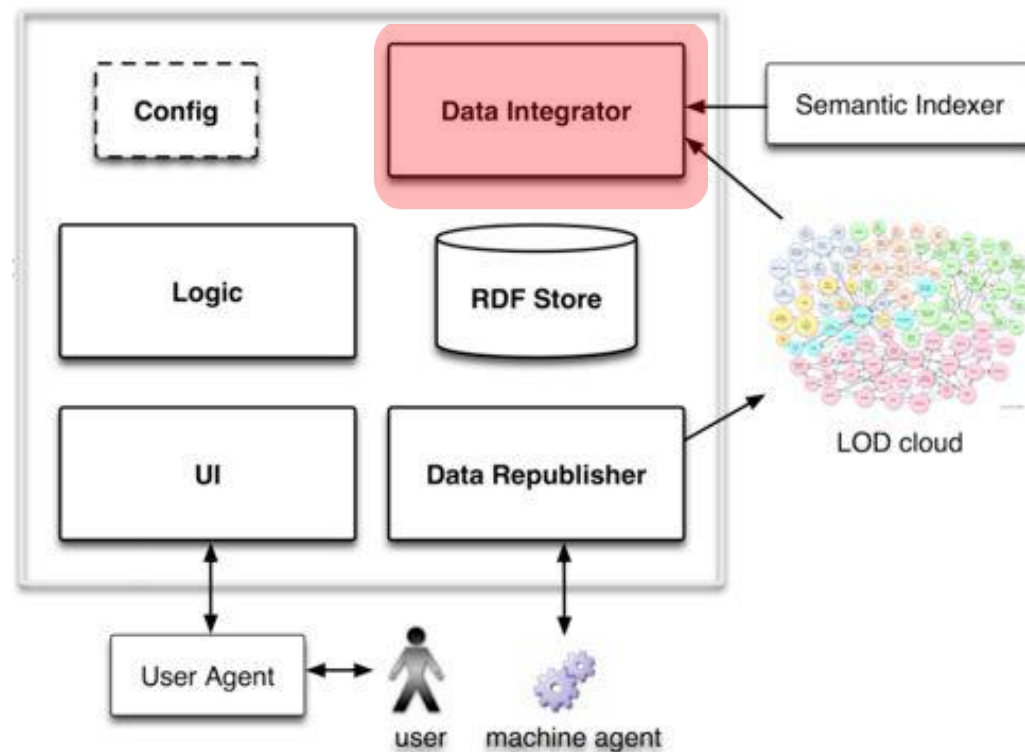
Knowledge Graph Driven (Web) Applications

- Required Components:
 - **Logic** (Controller) and
 - **User Interface**
 - (=Business Logic)
 - (not KG specific)



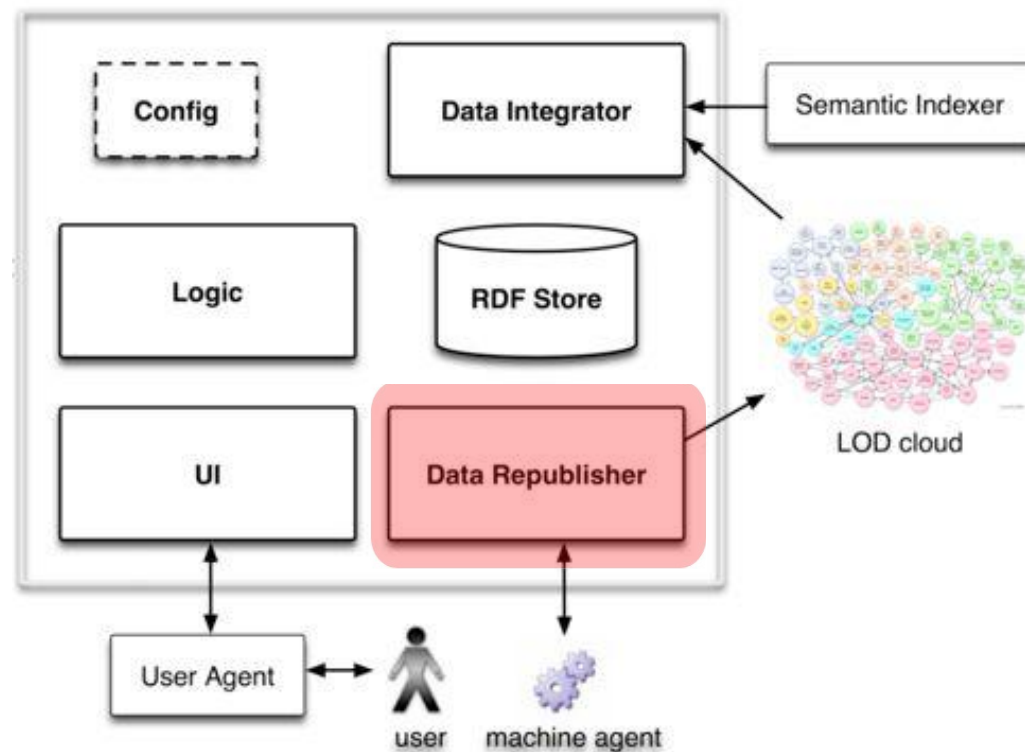
Knowledge Graph Driven (Web) Applications

- Required Components:
- **Data Integration component**
 - get data directly from Knowledge Graph or
 - via Semantic Indexer



Knowledge Graph Driven (Web) Applications

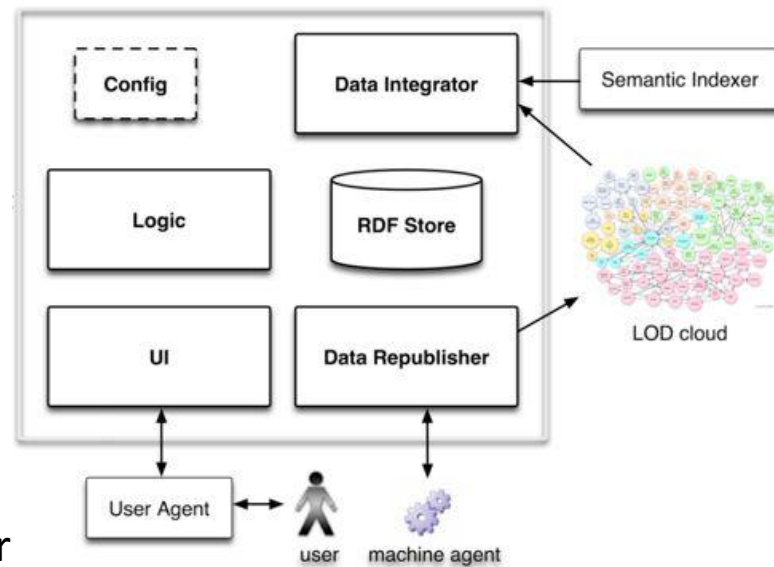
- Required Components:
- **Data Re-/Publishing component**
 - write back application dependent data into the Web of Data



Knowledge Graph Driven (Web) Applications

- Required Components:

- **Local RDF Store**
 - caching of results
 - permanent storage
- **Logic** (Controller) and
- **User Interface** (=Business Logic)
 - (not KG specific)
- **Data Integration component**
 - get data directly from Web of Data or
 - via Semantic Indexer
- **Data Re-/Publishing component**
 - write back application dependent data into the Web of Data



Knowledge Graph Driven (Web) Applications

- The easiest way is to make use of a suitable library:
 - SPARQL Javascript Library
http://www.thefigtrees.net/lee/blog/2006/04/sparql_calendar_demo_a_sparql.html
 - ARC for SPARQL (PHP)
<https://github.com/semsol/arc2/wiki>
 - dotNetRDF (C#)
<https://dotnetrdf.github.io/>
 - Jena/ARQ (Java)
<http://jena.apache.org/>
 - Sesame (Java)
<http://rdf4j.org/>
 - **RDFLib / SPARQL Wrapper (Python)**
<http://rdflib.github.io/sparqlwrapper/>

Knowledge Graph Driven (Web) Applications

- The easiest way is to make use of a suitable library:
 - **SPARQL Wrapper (Python)**
<http://rdflib.github.io/sparqlwrapper/>
- Access to Linked Data via **SPARQL endpoints**
 - Let's choose **DBpedia** (just for simplicity...)
<http://dbpedia.org/sparql>
- ...now we have to think of a simple example...

Knowledge Graph Programming Example

- **Example Application:**

- Build a simple application that looks for famous people having their birthdays **today(!)**, as e.g. **scientists**, based on **DBpedia**
- Create a **list of scientists**, whose birthday is today including some **additional information**, as e.g.
 - Year of Birth
 - Short description, picture (if available), etc.
- Let's create a simple (local) **web page** for the task (i.e. encode results in HTML), which can be displayed in the browser
- We use **Python** and the **SPARQL Wrapper library**

Prerequisites - (Manual) Data Analysis

- Choose a representative example:
 - E.g. **Joseph Fourier** from **DBpedia**

 Browse using ▾ Formats ▾

Faceted BrowserSparql Endpoint

About: Joseph Fourier

An Entity of Type : [scientist](#), from Named Graph : <http://dbpedia.org>, within Data Space : [dbpedia.org](#)

Jean-Baptiste Joseph Fourier - (/ˈfoʊriˌeɪ, -iər/; French: [fʁusʒe]; 21 March 1768 – 16 May 1830) was a French mathematician and physicist born in Auxerre and best known for initiating the investigation of Fourier series and their applications to problems of heat transfer and vibrations. The Fourier transform and Fourier's law are also named in his honour. Fourier is also generally credited with the discovery of the greenhouse effect.

Property	Value
dbpedia:abstract	- Jean Baptiste Joseph Fourier (* 21. März 1768 bei Auxerre; † 16. Mai 1830 in Paris) war ein französischer Mathematiker und Physiker. (de) - Jean-Baptiste Joseph Fourier - (/ˈfoʊriˌeɪ, -iər/; French: [fʁusʒe]; 21 March 1768 – 16 May 1830) was a French mathematician and physicist born in Auxerre and best known for initiating the investigation of Fourier series and their applications to problems of heat transfer and vibrations. The Fourier transform and Fourier's law are also named in his honour. Fourier is also generally credited with the discovery of the greenhouse effect. (en)
dbpedia:academicAdvisor	dbr:Joseph-Louis_Lagrange
dbpedia:almaMater	dbr:École_Normale
dbpedia:birthDate	1768-03-21 (xsd:date) 1768-3-21
dbpedia:birthPlace	dbr:Burgundy_(region) dbr:Kingdom_of_France

http://dbpedia.org/page/Joseph_Fourier

Prerequisites - (Manual) Data Analysis

- Data Analysis via **SPARQL**

- What kind of entities are you looking for?

- `?scientist rdf:type dbo:Scientist .`

- What information do you need?

- `?scientist dbo:birthDate ?birthdate .`

- `?scientist rdfs:label ?name .`

- `?scientist rdfs:comment ?description .`

- `OPTIONAL { ?scientist dbo:thumbnail ?thumbnail }`

- Any filter criteria?

- `(lang(?name)="en") && (lang(?description)="en")`

- `(SUBSTR(STR(?birthdate),6)="18-03")`

- More sophisticated: `(SUBSTR(STR(bif:curdate('')),6))`

Virtuoso triple
store builtin
function for
current date

[4]

Prerequisites - SPARQL Queries

[SPARQL query at dbpedia.org](https://dbpedia.org)

Virtuoso SPARQL Query Editor

Default Data Set Name (Graph IRI)

Query Text

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>

SELECT distinct ?birthdate ?thumbnail ?scientist ?name ?description WHERE {
?scientist rdf:type dbo:Scientist ;
          dbo:birthDate ?birthdate ;
          rdfs:label ?name ;
          rdfs:comment ?description
FILTER ((lang(?name)="en")&&(lang(?description)="en")&&
        (STRLEN(STR(?birthdate))>6)&&(SUBSTR(STR(?birthdate),6)=SUBSTR(STR(bif:curdate(''),6))) ) .
OPTIONAL { ?scientist dbo:thumbnail ?thumbnail . }
} ORDER BY ?birthdate
```

(Security restrictions of this server do not allow you to retrieve remote RDF data, see [details](#).)

Results Format:

Execution timeout: milliseconds (values less than 1000 are ignored)

Options:

- ☒ Strict checking of void variables
- ☐ Log debug info at the end of output (has no effect on some queries and output formats)
- ☐ Generate SPARQL compilation report (instead of executing the query)

(The result can only be sent back to browser, not saved on the server, see [details](#))

Python Code in a Collaborative Notebook

Scientific Birthdays Example

This is the python notebook example for lecture 3.4 Knowledge Graph Programming, of the OpenHPI lecture 'Knowledge Graphs 2020'.

Please make a copy of this notebook to try out your own adaptations via "File -> Save Copy in Drive"

First, we have to install the **spqrlwrapper** library before we can use it with the notebook.

```
[ ] !pip install -q spqrlwrapper #install SPARQLWrapper
```

We are going to use a few libraries:

- **datetime** for date formatting and interpretation
- **SPARQLWrapper** to execute SPARQL queries and to import the results into python

Thus, we will import them now.

```
[ ] from datetime import datetime
from SPARQLWrapper import SPARQLWrapper, JSON, XML, N3, RDF
```

We will use DBpedia (<http://dbpedia.org/sparql>) as our SPARQL endpoint

```
[ ] sparql = SPARQLWrapper("http://dbpedia.org/sparql") #determine SPARQL endpoint
```

Next comes the query example from the lecture and its execution

```
[ ] #SPARQL query to be executed
sparql.setQuery("""
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX dc: <http://purl.org/dc/elements/1.1/>

Select distinct ?birthdate ?thumbnail ?scientist ?name ?description WHERE {
?scientist rdf:type dbo:Scientist ;
dbo:birthDate ?birthdate ;
rdfs:label ?name ;
rdfs:comment ?description
FILTER ((lang(?name)="en")&&(lang(?description)="en")&&(STRLEN(STR(?birthdate))>6)&&(SUBSTR(STR(?birthdate),6)=SUBSTR(STR(bif:curdate('')),6))) .
OPTIONAL { ?scientist dbo:thumbnail ?thumbnail . }
} ORDER BY ?birthdate
""")

sparql.setReturnFormat(JSON) # Return format is JSON
results = sparql.query().convert() # execute SPARQL query and write result to "results"
```

[Link to collaborative Notebook](#)

Knowledge Graph Programming Example

Scientific Birthdays of Thursday January 30

- 1720



[Charles De Geer](#), Baron Charles de Geer (the family is usually known as De Geer with a capitalized "De"; Finspång in Risinge 30 January 1720 – Stockholm 7 March 1778) was a Swedish industrialist and entomologist.
- 1883



[Andrew Robertson \(engineer\)](#), He was the son of a marine engineer. He was a graduate of the University of Manchester, with a first-class honours degree
- 1898



[George F.C. Griss](#), George François Cornelis Griss (30 January 1898, Amsterdam – 2 August 1953, Blaricum), usually cited as G. F. C. Griss, was a Dutch mathematician and philosopher, who was occupied with hegelian idealism and Brouwers intuitionism and stated a negationless mathematics.
- 1898



[Russell Ohl](#), Russell Shoemaker Ohl (January 30, 1898 – March 20, 1987) was an American engineer who is generally recognized for patenting the modern solar cell (US Patent 2402662, "Light sensitive device").Ohl was a notable semiconductor researcher prior to the invention of the transistor. He was also known as R.S. Ohl.
- 1899



[Max Theiler](#), Max Theiler (30 January 1899 – 11 August 1972) was a South African-American virologist and doctor. He was awarded the Nobel Prize in Physiology or Medicine in 1951 for developing a vaccine against yellow fever in 1937, becoming the first African-born Nobel laureate.

[Evelyn Hutchinson](#), George Evelyn Hutchinson (January 30, 1903 – May 17, 1991), was an American ecologist sometimes

You may also try out Wikidata

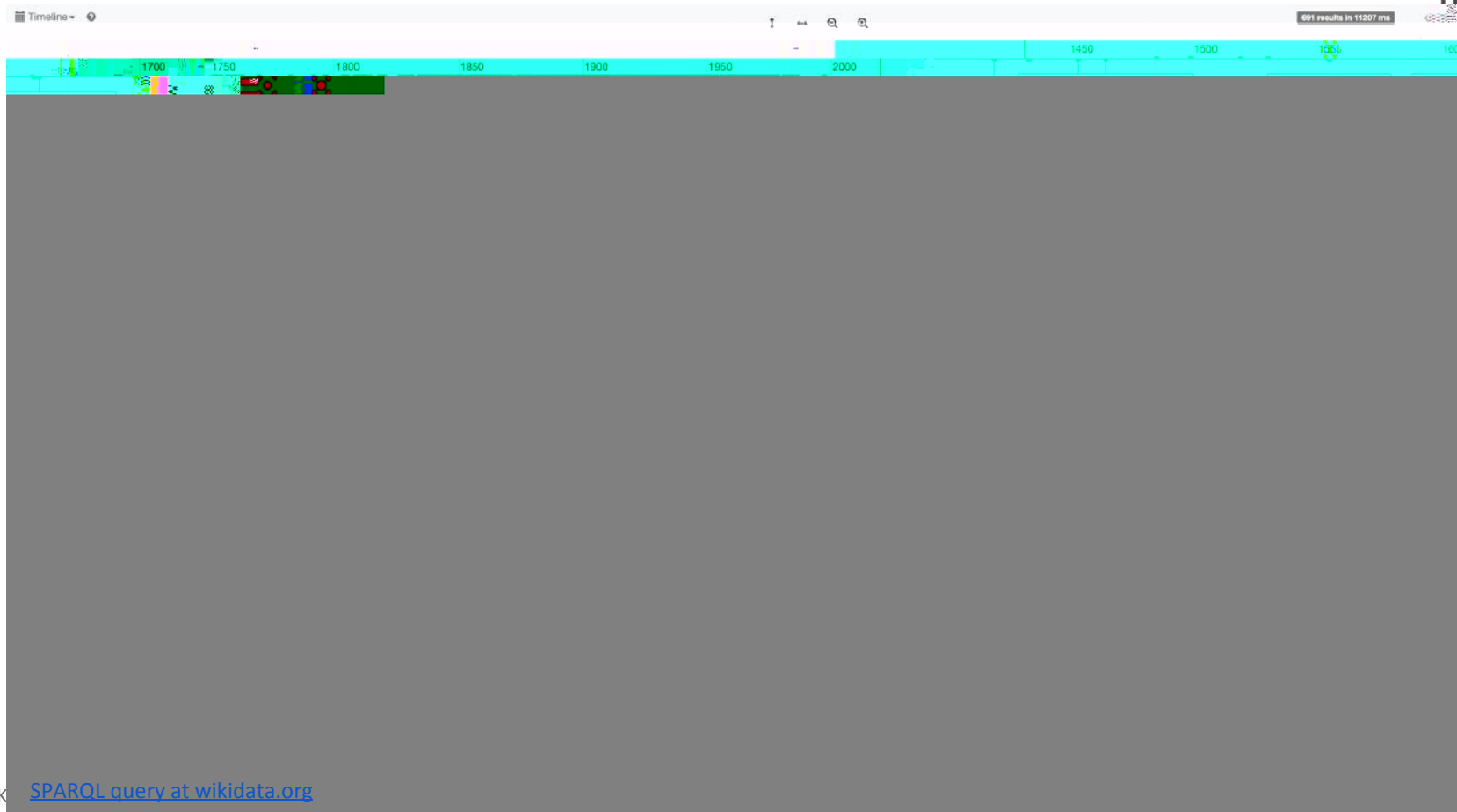
```

PREFIX wdt: <http://www.wikidata.org/prop/direct/> # Properties
PREFIX wd: <http://www.wikidata.org/entity/>      # Entities

#defaultView:Timeline
SELECT distinct ?birthdate ?thumbnail ?author ?authorLabel ?authorDescription
WHERE {
  BIND(MONTH(NOW()) AS ?nowMonth) # NOW() is current date, create new variable for current month
  BIND(DAY(NOW()) AS ?nowDay)     # create new variable for current day
  ?author wdt:P106/wdt:P279* wd:Q901 ;      # (P106) occupation (Q901) scientist
      wdt:P569 ?birthdate                # (P569) birthdate
  # compare birthdate and current date
  FILTER (MONTH(?birthdate) = ?nowMonth && DAY(?birthdate) = ?nowDay) .
  OPTIONAL { ?author wdt:P18 ?thumbnail . } # (P18) image
  SERVICE wikibase:label
    { bd:serviceParam wikibase:language "en, de, fr, it" } # provide labels in English
}
ORDER BY ?birthdate

```


You may also try out Wikidata



Knowledge Graph Visualization

Next Lecture...

Picture References:

- [1] Utagawa Kuniyoshi, The three tiles in Takiyasha the Witch and the Skeleton Spectre, [Public Domain]
[https://commons.wikimedia.org/wiki/File:Takiyasha the Witch and the Skeleton Spectre.jpg](https://commons.wikimedia.org/wiki/File:Takiyasha_the_Witch_and_the_Skeleton_Spectre.jpg)